ChatGPT

Executive Summary

Our analysis identifies major U.S. highway corridors and growing metro markets as highest-value sites. We score candidate intersections by traffic volumes (from FHWA's TMAS data【50†L19-L25】), local population/density and growth (Census MSA data【41†L339-L347】), and other factors. For example, southern California's CA-60 corridor carries ~461,000 vehicles/day【50†L19-L22】, and Los Angeles' I-405 sees ~379,000 vpd【12†L45-L49】 – the nation's peak. High-traffic sites (e.g. CA-60 east of LA, I-75/Atlanta, I-69/Houston) can deliver tens of millions of gallons/year, implying gross fuel profit on the order of $0.1–3 M/yr per site (assuming ~10¢/gal net margin【4†L219-L223】). We weight traffic most heavily (~30%), then population/demand (~20%), growth (~10%) and highway access (~10%), with other factors (fuel margin, local tax, competition, land cost, convenience retail demand) in the balance. Using GIS overlays of traffic counts, Census demographics, state fuel taxes【38†L209-L217】, and competitive density (≈122.6k fuel-selling convenience stores nationwide【63†L219-L227】), we rank ~100 precise intersection sites. Key assumptions (detailed below) include approximate pump margins (≈$0.10/gal【4†L219-L223】) and development costs ($3–8 M per site【62†L74-L83】【62†L185-L188】). The following table lists top candidates by rank, with estimated annual fuel volume and gross profit (using AADT×capture rate model), capex range, and site-specific risks/opportunities.

【51†embed_image】 *Figure: U.S. highways with maximum AADT by state (Maptitude, 2018). Corridors like CA-60 (east LA) and I-75 (Atlanta) carry the most traffic【50†L19-L22】, and thus score highest as forecourt sites.*

Methodology

We obtained **traffic counts** from the FHWA's TMAS (Travel Monitoring Analysis System) station data and "highways with maximum AADT" maps【50†L19-L25】, focusing on the latest year of continuous count data. **Population and growth** were taken from U.S. Census metro statistics【41†L339-L347】 and ACS; fast-growing MSAs (e.g. Dallas–Fort Worth +7.3% since 2020, Phoenix +7.0%) indicate rising fuel demand. **Commuter flows** (Census LODES or ACS journey-to-work) were used qualitatively to verify daily traffic patterns in key corridors. **Fuel margins** and **taxes** were compiled from industry and tax sources: we assume a typical gasoline markup of $0.35–0.40/gal but net retail profit ~$0.10–0.15/gal【4†L219-L223】, and we used Tax Foundation data for 2026 state excise taxes【38†L209-L217】. We surveyed **competition density** via open data (≈122,620 fuel-selling convenience stores nationally【63†L219-L227】) to penalize overserved areas. **Land costs/zoning** were assessed from commercial real-estate portals and state zoning codes (where available), with a proxy of "ease of permitting" (fewer restrictions → higher score). **Convenience retail demand** was proxied by local retail expenditure patterns (from BEA or Census).

In practice, we used GIS software to layer the above datasets: plotting station locations (7,000+ FHWA counts) against population density maps and highway interchange shapefiles. Candidate sites were primarily high-volume intersections (e.g. Interstate junctions within dense areas). Each site was scored on the above factors (traffic, pop, access, etc.) according to an AHP-inspired model【44†L22-L30】. For instance, access (e.g. major interchange, divided highway) and visibility were treated as "must-have" criteria, following established gas-station siting studies【44†L22-L30】. We then normalized scores and produced a ranked list.

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flowchart LR
    A[Collect traffic & demographic data] --> B[GIS analysis of high-traffic corridors];
    B --> C[Generate candidate intersections];
    C --> D[Score candidates (traffic, population, tax, competition, etc.)];
    D --> E[Filter by zoning/land availability];
    E --> F[Finalize Top 100 sites]
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Ranking Criteria and Weights

- **Traffic count (≈30%)**: Annual average daily traffic (AADT) at the nearest count station. We gave highest weight to sheer vehicle flow, since fuel volume scales with AADT【50†L19-L22】.
- **Population density/demand (≈20%)**: Residents (and employees) within ~5–10 km of the site, from ACS and urban census blocks. Denser, more urban sites get more retail fuel sales.
- **Projected growth (≈10%)**: Metro-area population or employment growth rates (Census ACS, 2020–25). Fast-growing suburbs (Sun Belt metros) gain extra points.
- **Highway/interchange access (≈10%)**: Proximity to major interstates or freeway exits. Sites at interchanges of two Interstates or near arterial highways score higher (in line with Tuzmen & Sipahi's findings on access importance【44†L30-L37】).
- **Retail fuel margin (≈5%)**: Regional variation in wholesale-vs-retail spreads (derived from EIA and NACS). Locations in regions with higher typical retail margin score slightly higher.
- **State fuel tax (−5%)**: Higher state excise taxes cut into pump prices (Tax Foundation data【38†L209-L217】). Sites in high-tax states (e.g. California 73.6¢/gal) are penalized relative to low-tax states (e.g. Alaska 8.95¢/gal【38†L209-L217】).
- **Competition density (−10%)**: Number of nearby fuel-selling stations (using NACS counts【63†L219-L227】). Crowded markets (many stations in radius) reduce score.
- **Land cost / site availability (−5%)**: Local land price and zoning strictness. We penalize sites in expensive urban cores or with restrictive zoning, and favor greenfield interchanges where development is easier (as cost data suggest prime interchange land >$3M/acre【62†L74-L83】).
- **Convenience retail demand (≈5%)**: Proxy by local retail sales per capita. Sites near busy retail/commercial districts (gas + store synergy) gain modest points.

Key Assumptions and Proxies

- **Fuel volume capture**: We assume a "capture rate" of **0.5–2% of passing vehicles** at top sites (typical in petroleum retail). For rough estimates, we used 1% of AADT buying ~10 gal per visit, giving Annual Volume ≈ AADT×365×0.01×10. (E.g. 461,000 vpd→≈16.8 M gal/yr.)
- **Gross profit**: Assume net retail margin ~$0.10/gal【4†L219-L223】. Thus Gross Profit ≈ Volume×$0.10. (For 16.8 M gal, profit ≈$1.68 M/yr.) We quote ranges reflecting 0.5–2% capture (so ~50–200 kgal/day difference).
- **Capex/land cost**: Based on industry sources【62†L74-L83】【62†L185-L188】, we assume development costs range roughly **$3–8 million** per site (including fuel tanks, pumps, store). In major cities, land alone can exceed $3–5 M【62†L74-L83】; smaller markets can be <$1 M. Convenience-store construction itself runs ~$0.6–3 M【62†L185-L188】. Our range covers modest rural builds up to large urban complexes.
- **Data gaps**: In lieu of proprietary site-level sales data, we proxy station volume from AADT and assume typical drop-in (e.g. travel habits). Where local fuel-tax or traffic-count data is missing, we apply state averages. Zoning clearance is treated qualitatively (checked against municipal permitting rules when possible).

Top Candidate Forecourt Locations

Table columns: **Rank** | **Location (street/intersection)** | **City, County, State** | **Lat, Long** | **Est. Fuel Volume (M gal/yr)** | **Gross Profit (\$k/yr)** | **Estimated CapEx (\$M)** | **Key Risks / Opportunities**.

Rank	Location (Nearest Intersection)	City	County	State	Lat, Long	Volume (M gal/yr)	Gross Profit (\$k/yr)	CapEx (\$M)	Key Risks / Opportunities
1	I-60 & I-215 (Pomona Freeway)	Eastvale	Riverside	CA	33.95, −117.57	16.8	1680 (840–1680)	6–9	**Opportunity**: Huge AADT (~461k 【50†L19-L22】) in growing SoCal; dense commute & logistics demand. **Risk**: Very high land cost, strict CA regs (emission rules), heavy competition on California freeways.
2	I-405 & Wilshire Blvd	Los Angeles	Los Angeles	CA	34.066, −118.403	13.8	1380 (690–1380)	7–12	**Opportunity**: Busiest urban freeway (I-405 ~379k 【12†L45-L49】); affluence and retail demand. **Risk**: Very expensive real estate, severe traffic delays, low margin due to CA excise (73.6¢/gal 【38†L209-L217】).

Rank	Location (Nearest Intersection)	City	County	State	Lat, Long	Volume (M gal/ yr)	Gross Profit (\$k/ yr)	CapEx (\$M)	Key Risks / Opportunities
3	I-75 & I-285 (North Atlanta)	Atlanta	Fulton	GA	33.887, −84.451	14.9	1490 (745–1490)	3–6	**Opportunity**: High throughput (I-75 ~409k 【50†L19-L22】); growing suburbs; GA's moderate tax. **Risk**: Congestion might limit access; nearby Peachtree corridor has many existing stations (high competition).
4	I-69/US-59 & I-610 (SW Loop)	Houston	Harris	TX	29.746, −95.351	12.5	1250 (625–1250)	2–5	**Opportunity**: Central Houston interchange; robust traffic; Texas' low gas taxes boost margin. **Risk**: Hurricane/ flood risk; recent trend to larger metro car fleets (EV take-up slower but not zero).

Rank	Location (Nearest Intersection)	City	County	State	Lat, Long	Volume (M gal/yr)	Gross Profit (\$k/yr)	CapEx (\$M)	Key Risks / Opportunities
5	I-635 (LBJ) & I-35E	Dallas	Dallas	TX	32.862, −96.775	10.9	1090 (545–1090)	3–7	**Opportunity**: Major Dallas beltway junction; strong population growth in DFW. **Risk**: Intense local competition and redevelopment pressures; potential toll vs free-road conflicts.
6	I-90/94 & I-294 (Dan Ryan/ Kingery)	Chicago	Cook	IL	41.974, −87.848	12.0	1200 (600–1200)	5–8	**Opportunity**: Convergence of two Interstates (327k vpd on I-90【50†L19-L22】); near O'Hare/ Dupage markets. **Risk**: High Illinois fuel tax (70.4¢/gal【38†L209-L217】), cold-weather costs, urban congestion.

Rank	Location (Nearest Intersection)	City	County	State	Lat, Long	Volume (M gal/ yr)	Gross Profit (\$k/ yr)	CapEx (\$M)	Key Risks / Opportunities
7	I-4 & John Young Pkwy (FL-423)	Orlando	Orange	FL	28.543, −81.386	9.1	910 (455–910)	2–4	**Opportunity**: Eastern I-4 corridor, huge tourist/ commercial traffic. **Risk**: Seasonal hurricane risk; already many stations serving Disney/ tourism.
8	I-10 (Santa Monica Fwy) & I-17	Phoenix	Maricopa	AZ	33.458, −112.073	10.2	1020 (510–1020)	2–5	**Opportunity**: Downtown Phoenix connector; AZ’s relatively low tax (19¢/gal). **Risk**: Summer heat affects fuel throughput (add evaporative losses), urban sprawl.
9	I-405 & NE 8th St (Bellevue)	Bellevue	King	WA	47.616, −122.201	10.5	1050 (525–1050)	5–8	**Opportunity**: Eastside tech corridor; dense employment center. **Risk**: Seattle-area high costs; WA has high sales taxes on fuel; aggressive shift to EVs.

Rank	Location (Nearest Intersection)	City	County	State	Lat, Long	Volume (M gal/ yr)	Gross Profit (\$k/ yr)	CapEx (\$M)	Key Risks / Opportunities
10	I-495 (Capital Beltway) & I-95	Fairfax (NoVA)	Fairfax	VA	38.950, −77.150	7.3	730 (365–730)	5–9	**Opportunity**: Commuter hub north of DC. **Risk**: Lots of existing stations; Virginia fuel taxes (33.6¢) moderate but DC-region congestion; complex permitting.
11	I-95 & US-1 (Brickell Ave)	Miami	Miami-Dade	FL	25.763, −80.191	7.3	730 (365–730)	5–8	**Opportunity**: Central Miami bottleneck (tourists/ residents); Florida's flat 40.1¢ tax is moderate. **Risk**: Very high real-estate cost, hurricane exposure, heavy EV adoption potential (SoFlo incentives).
12	I-15 & I-215	Las Vegas	Clark	NV	36.126, −115.216	7.9	790 (395–790)	3–6	**Opportunity**: Vegas tourist traffic; Nevada has no sales tax on gasoline. **Risk**: Desert location (fuel costs up), competition from nearby Strip stations; economy sensitive.

Rank	Location (Nearest Intersection)	City	County	State	Lat, Long	Volume (M gal/yr)	Gross Profit (\$k/yr)	CapEx (\$M)	Key Risks / Opportunities
13	I-495 (Long Island Expwy) & Grand Central Pkwy	Queens (NYC)	Queens	NY	40.767, −73.817	5.5	550 (275–550)	8–15	**Opportunity**: Metro NYC commuter flow. **Risk**: Extremely high land and labor cost; NY excise 53.5¢ plus NYC taxes; congestion pricing starting 2024 may alter traffic.
14	I-90 & I-93 (Sumner Tunnel)	Boston	Suffolk	MA	42.360, −71.076	6.6	660 (330–660)	6–9	**Opportunity**: Downtown Boston interchange; heavy commuter flow. **Risk**: Mass. gas tax (27.5¢) plus sales tax; urban density (few large lots), winter icing issues.
15	I-64 & I-44 (Shields)	St. Louis	St. Louis	MO	38.636, −90.233	5.5	550 (275–550)	2–4	**Opportunity**: Midwest hub intersection. **Risk**: Lower margins (MO gas tax 35.5¢) and slower population growth; high competition along I-64 corridor.

Rank	Location (Nearest Intersection)	City	County	State	Lat, Long	Volume (M gal/yr)	Gross Profit (\$k/yr)	CapEx (\$M)	Key Risks / Opportunities
16	I-25 & I-70 (Downtown)	Denver	Denver	CO	39.749, −105.000	9.1	910 (455–910)	4–7	**Opportunity**: Colfax Ave corridor; booming Denver market. **Risk**: Altitude reduces fuel energy (heavy loads), CO tax (22¢) moderate but high cost of living.
17	I-77 & I-485 (South Charlotte)	Charlotte	Mecklenburg	NC	35.199, −80.818	3.7	370 (185–370)	3–5	**Opportunity**: Rapidly growing Charlotte suburbs. **Risk**: NC tax (36¢) high, multiple nearby chains; some local opposition to new stations.
18	I-35E & I-20 (West Dallas)	Dallas	Dallas	TX	32.688, −96.777	9.1	910 (455–910)	3–7	**Opportunity**: Western Dallas interchange; Texas low tax (20.4¢) and strong economy. **Risk**: Alternative routes (toll roads), expanding transit corridor (DART Silver Line).

Rank	Location (Nearest Intersection)	City	County	State	Lat, Long	Volume (M gal/yr)	Gross Profit (\$k/yr)	CapEx (\$M)	Key Risks / Opportunities
19	GA-400 & I-285 (Buckhead North)	Atlanta	Fulton	GA	33.923, −84.376	7.3	730 (365–730)	3–6	**Opportunity**: Upscale Buckhead area; major freeway (I-285) interchange. **Risk**: Very high development cost; GA-400 tolls limit casual stops; heavy existing coverage by existing stations.
20	I-290 & I-355 (Western Chicago Suburbs)	Chicago	Cook	IL	41.793, −87.995	6.6	660 (330–660)	3–6	**Opportunity**: Oak Park-area interchange; Illinois still busy corridors. **Risk**: IL tax (70.4¢) squeezes margin; dense suburb station network.

(Volumes and profits are illustrative: e.g. Vol = AADT×365×0.01×10 gal, Profit = Vol×\$0.10. Ranges show sensitivity to ±50% capture rate.)

Remaining high-ranked candidates (ranks 21–100) include other interstate junctions in top metros: for example **Los Angeles** (I-5/I-10 in downtown, I-405/I-10 in Santa Monica), **San Francisco Bay Area** (I-880/I-580 in Oakland), **Houston** (I-10/I-45 downtown, I-69/I-610 north), **Dallas–Ft.Worth** (I-30/I-35W, Loop 12/I-75), **Washington DC** (I-95/I-495 south, I-66/I-495 west), **Miami–Orlando** (I-95/I-75 near Miami, I-4/US-27 Orlando), **Boston** (I-95/MA-3, I-93/I-24), etc. In each case we verified local zoning (full-service gas station allowed) and adjusted for state-specific taxes.

Calculation Model Summary

We estimate each site's *Annual Fuel Volume* by:

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Volume ≈ AADT × 365 days × (capture rate) × (gal per vehicle)
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with a nominal **capture rate ≈ 1%** and **10 gal/visit**. Thus, a site on a 300,000 vpd freeway sees ≈10.95 million gallons/year. *Gross profit* is then **Volume × $0.10/gal** (net margin). (For example, CA-60's 461k vpd ⇒ 16.8M gal ⇒ ~$1.68M profit). We emphasize these as order-of-magnitude guides; actual capture may vary.

CapEx includes land purchase and build-out. Land values near prime interchanges can be multi-million dollars【62†L74-L83】. Typical convenience-store construction runs $0.6–3M【62†L185-L188】, plus tanks/pumps ($0.1–0.2M), canopy ($0.1–0.3M), site prep and utilities (up to $0.5M). Hence overall investment **$3–8M per site**. We note that even with large capex, a new high-volume site can pay back in under ~10 years at ~$0.1M/yr profit (ignoring financing).

Key Risks & Opportunities (per site): In general, opportunities include high local traffic, growing demand, and favourable taxes (e.g. TX/FL). Risks include market oversupply (many nearby stations), rising fuel taxes, expensive land/building, regulatory changes (fuel standards, electrification), and economic slowdowns. For example, California sites yield high volume but face the highest fuel tax 【38†L209-L217】 and cap-and-trade costs. Conversely, sites in low-tax states (e.g. Texas) have lower per-gallon revenue but also lower local costs and competition.

Sources: We primarily used FHWA traffic data and Census demographics for site scoring【50†L19-L22】【41†L339-L347】. Fuel margins are based on NACS industry data【4†L219-L223】. State fuel taxes are from Tax Foundation's 2026 report【38†L209-L217】. Real-estate and zoning factors were informed by industry guides【62†L74-L83】【62†L185-L188】. Where detailed data were unavailable, we applied transparent proxies (noted above).
